

SAI PRANAY TADAKAMALLA

B.Tech, Artificial Intelligence & Machine Learning | Founder & CEO, 221B Labs Private Limited
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ABOUT

Final-year undergraduate in Artificial Intelligence and Machine Learning at Malla Reddy University. Research work covers exploration in agentic AI, cooperative multi-agent robotics, software testing, speech-based reading support for under-served languages and memory limits of predictive models, with five preprints. Founder and CEO of 221B Labs Private Limited, a software company that runs campus event ticketing and is building a speech-based reading platform for Indian classrooms. Interests: reinforcement learning, agentic AI, quantum-inspired algorithms and speech technology for Indian languages.

EDUCATION

B.Tech, Artificial Intelligence & Machine Learning — Malla Reddy University, Hyderabad | CGPA 8.85 / 10 2023 – 2027 (exp.)
Class XII — Geetanjali Junior College, Nalgonda | 86.4% 2023
Class X — Kendriya Vidyalaya, Nalgonda | 76.8% 2021

RESEARCH EXPERIENCE

Undergraduate Researcher — Dept. of AI & ML, Malla Reddy University | Guide: Manasa Chandupatla 2026 – Present

- Major project: *Tunnelling Through Deception*, a quantum-inspired exploration plug-in for escaping deceptive reward traps; in a 40-seed pre-registered study, classical and quantum variants scored 0.875 and 0.877 against a 0.494 baseline ($p < 0.0001$).
- Built a reproducible, tested Python codebase and live comparative demo (github.com/PranayTadakamalla/final-project); manuscript in preparation for IEEE submission.

PUBLICATIONS (PREPRINTS)

- Tadakamalla, S. P.**, Prakash, S. M. R., & Asad, S. M. (2026). *Predictive State Is Not Explanatory State: An Exponential Memory Separation Between Forecasting and Counterfactual Auditing*. Preprint. [Code & paper \(GitHub\)](#)
- Tadakamalla, S. P.**, & Asad, S. M. (2026). *Heard in Their Own Voice: A Language-Agnostic Framework for AI-Assisted Pronunciation and Reading Support in the World's Under-Served Languages*. SSRN. ssrn.com/abstract=7488098
- Tadakamalla, S. P.**, & Terala, S. (2026). *The Fullness That Remains: Quantum Non-Locality, Mathematical Infinity, and the Search for an Undivided Whole*. SSRN. [doi:10.2139/ssrn.7350098](https://doi.org/10.2139/ssrn.7350098)
- Tadakamalla, S. P.**, & Edukoju, G. S. (2025). *From Pixels to Paths: Cooperative AI for Autonomous Agricultural Navigation*. SSRN. [doi:10.2139/ssrn.5935575](https://doi.org/10.2139/ssrn.5935575)
- Chandupatla, M., **Tadakamalla, S. P.**, Prakash, S. M. R., & Y, D. (2025). *A Unified Framework for End-to-End Testing in E-Commerce Applications: Towards AI-Driven Resilience and Automation*. SSRN. [doi:10.2139/ssrn.5602510](https://doi.org/10.2139/ssrn.5602510)

EXPERIENCE

Founder & CEO — 221B Labs Private Limited, Hyderabad | Incubated at Malla Reddy University Mar 2026 – Present

- Lead strategy, product, client delivery and hiring; run an engineering internship programme.
- Festora**: built the company's event ticketing platform, used to ticket 5,000+ students; official ticketing and community partner of Malla Reddy University and DataForge, a Microsoft developer community in Hyderabad. Delivered the MALCON '26 registration and QR check-in system for 700+ participants, covering entry and two-day meal passes.
- Pathana Sakthi** (Speech AI, TTS, G2P, Android): offline read-along tutor for Classes 1–10 in Telugu, English and Hindi, aligned to the SCERT curriculum, with teacher, parent and admin dashboards; building kid-calibrated Indian voices, a language model and G2P conversion, with word-level pronunciation scoring through CTC forced alignment.
- Pathana Sakthi pilot with SCERT stakeholders: 2–3 government schools, 150–300 students, targeting 75%+ read-along accuracy.

Project Manager Intern — Pramila Foundation Jan 2025 – Jun 2025

- Led development of the organisation's website and mobile app; feature and UX changes raised user engagement by 85%.

PROJECTS

CTI-IDS — Threat Intelligence & Intrusion Detection | BERT, LSTM, CNN, Next.js, Python | [GitHub](#)

- Detects phishing and malware from email text (BERT), network packet sequences (LSTM) and screenshots (CNN); 92% accuracy for BERT; enriched with VirusTotal, URLhaus and Shodan feeds.

Bazinga Labs — LLM Code Generation Platform | React.js, Node.js, LLMs

- Generates code in Python, Java, TypeScript, React, AngularJS and Node.js through a secure, scalable API backend.

AADHARVA — AI Digital Hub for Rural Advancement | Next.js, Python, Google APIs | [GitHub](#)

- Multilingual conversational assistant that guides rural users through digital services.

CertChain — Blockchain Credential Verification | Solidity, ethers.js, IPFS, PostgreSQL, Python

- QR-coded certificates anchored on a Solidity registry via IPFS; designed H-PCAE (PCA + autoencoder + entropy selection) to compress 256-dimensional credential records into a 32-byte SHA-256 fingerprint stored on-chain.

ACHIEVEMENTS & LEADERSHIP

2nd Place, Pitch Arena - Ideathon 4.0 — Malla Reddy University Aug 2026

- Pitched Pathana Sakthi; advanced to Round 2 of the National Entrepreneurship Challenge 2026 (E-Cell, IIT Bombay).

Vice-President — Microsoft Learn Student Chapter, Malla Reddy University Apr 2025 – Oct 2025

- Led a 20-member team that ran 10+ workshops on AI, cloud and software development, reaching 5,000+ students.

SKILLS, CERTIFICATIONS & LANGUAGES

AI/ML: Machine Learning, Reinforcement Learning, NLP, LLMs, Prompt Engineering, Speech Recognition, Experimental Design
Programming & Web: Python, Java, JavaScript, TypeScript, React.js, Next.js, Node.js, Express.js, Django, REST APIs, HTML/CSS
Data, Cloud & Tools: MongoDB, MySQL, AWS, Google Cloud Platform, CI/CD, Git, GitHub, Postman
Certifications: Elements of AI (University of Helsinki), Reinforcement Learning (NPTEL), B2 English (Cambridge University)
Languages: English (professional, CEFR B2), Hindi (full professional, C1), Telugu (native)